

# OSSP – SKILL

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2520030332

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## WEEK1

```
root@Hansi: ~/OSSP
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execl("/bin/ls", "ls", NULL);
    }
    else if (pid > 0)
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);

        printf("Child process completed.\n");
    }
    else
    {
        perror("fork failed");
    }

    return 0;
}
```

```
root@Hansi: ~/OSSP
root@Hansi:~# cd ~/OSSP
root@Hansi:~/OSSP# cd ~/OSSP/2520030332_Skill
-bash: cd: /root/OSSP/2520030332_Skill: No such file or directory
root@Hansi:~/OSSP# pwd
/root/OSSP
root@Hansi:~/OSSP# nano skill1.c
root@Hansi:~/OSSP# gcc skill1.c -o skill1
root@Hansi:~/OSSP# ./skill1
Parent Process
Parent PID: 522
Child Process
Child PID: 523
Parent PID: 522
2520030332_Practical skill1 skill1.c
Child process completed.
root@Hansi:~/OSSP#
```

## WEEK2

```
root@Hansi: ~/OSSP
GNU nano 7.2
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}
```

```
root@Hansi: ~/OSSP
root@Hansi:~/OSSP# nano skill2.c
root@Hansi:~/OSSP# gcc skill2.c -o skill2
root@Hansi:~/OSSP# ./skill2
myshell> hello world
You entered: hello world
myshell> ls -l
You entered: ls -l
myshell> exit
Exiting shell...
root@Hansi:~/OSSP#
```

## WEEK3

```
root@Hansi: ~/OSSP
GNU nano 7.2 skill3.c *
#include <stdio.h>
#include <string.h>

int main()
{
    char history[5][100];
    char command[100];
    int count = 0;
    int i;

    printf("Simple Shell History\n");

    while (count < 5)
    {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "history") == 0)
        {
            printf("\nCommand History:\n");

            for (i = 0; i < count; i++)
            {
                printf("%d %s\n", i + 1, history[i]);
            }

            continue;
        }

        if (strcmp(command, "exit") == 0)
        {
            break;
        }

        strcpy(history[count], command);
        count++;
    }

    return 0;
}
```

```
root@Hansi: ~/OSSP
root@Hansi:~/OSSP# nano skill3.c
root@Hansi:~/OSSP# gcc skill3.c -o skill3
root@Hansi:~/OSSP# ./skill3
Simple Shell History
myshell> ls
myshell> pwd
myshell> data
myshell> history

Command History:
1  ls
2  pwd
3  data
myshell> █
```